



**Generation Interconnection
Impact Study Report
for
Queue Project AE1-168
VICTORIA-MARTIN 115 KV
90 MW Capacity / 150 MW Energy**

October, 2019

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement between Dickerson Creek Solar LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Nottoway County, Virginia. The installed facilities will have a total capability of 150 MW with 90 MW of this output being

recognized by PJM as Capacity. The proposed in-service date for this project is 9/01/2021. This study does not imply a TO commitment to this in-service date.

Queue Number	AE1-168
Project Name	VICTORIA-MARTIN 115 KV
Interconnection Customer	Dickerson Creek Solar LLC
State	Virginia
County	Nottoway
Transmission Owner	Dominion
MFO	150
MWE	150
MWC	90
Fuel	Solar
Basecase Study Year	2022

3.1 Point of Interconnection

AE1-168 will interconnect with the ITO transmission system via a new three breaker ring bus switching station that connects on the Victoria DP Tap to Martin DP 115 kV line #1034. See one line in Attachment 1.

3.2 Cost Summary

The AE1-168 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$1,550,000
Direct Connection Network Upgrade	\$5,500,000
Non Direct Connection Network Upgrades	\$ 800,000
Total Costs	\$7,850,000

In addition, the AE1-168 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$94,823,825

The System Impact Study provides refined and comprehensive estimates of cost responsibility and construction lead times for new facilities and system upgrades. Facilities Studies will include, commensurate with the degree of engineering specificity as provided in the Facilities Study Agreement, good faith estimates of the cost, determined in accordance with Section 217 of the Tariff,

(a) to be charged to each affected New Service Customer for the Facilities and System Upgrades that are necessary to accommodate this queue project;

(b) the time required to complete detailed design and construction of the facilities and upgrades; and

(c) a description of any site-specific environmental issues or requirements that could reasonably be anticipated to affect the cost or time required to complete construction of such facilities and upgrades.

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

4 Transmission Owner Scope of Work

4.1 Attachment Facilities

Generation Substation: Install metering and associated protection equipment. Estimated Cost \$550,000.

Transmission: Construct approximately one span of 115 kV Attachment line between the generation substation and a new AE1-168 Switching Station. The estimated cost for this work is \$1,000,000.

The estimated total cost of the Attachment Facilities is \$1,550,000. It is estimated to take 18-24 months to complete this work. These preliminary cost estimates are based on typical engineering costs. A more detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the generation substation during the Facility Study phase. The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Generator Substation	\$ 550,000
Transmission	\$1,000,000
Total Attachment Facility Costs	\$1,550,000

4.2 Direct Connection Cost Estimate

Substation: Establish the new 115 kV AE1-168 Switching Substation (interconnection substation). The estimated cost of this work scope is \$5,500,000. It is estimated to take 24-36 months to complete this work.

4.3 Non-Direct Connection Cost Estimate

Transmission: Install transmission structure in-line with transmission line to allow the proposed interconnection switching station to be interconnected with the transmission system. Estimated cost is \$800,000 and is estimated to take 24-30 months to complete.

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

5 Interconnection Customer Requirements

5.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion's "Dominion Energy Electric Transmission Generator Interconnection Requirements" documented in Dominion's Facility Interconnection Requirements "Exhibit C" located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

5.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC's GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion's "Dominion's Facility Interconnection Requirements" document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

5.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

6 Revenue Metering and SCADA Requirements

6.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

6.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

6.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

7 Network Impacts

The Queue Project AE1-168 was evaluated as a 150.0 MW (Capacity 90.0 MW) injection on the Victoria DP Tap to Martin DP 115 kV line #1034 in the Dominion area. Project AE1-168 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-168 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

8 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
3518714	314559	3CAROLNA	DVP	314561	6CAROLNA	DVP	1	DVP_P1-3: 6CLUBHSE-TX#1	single	239.89	99.4	102.96	AC	8.49
3518602	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	Base Case	single	176.81	97.33	100.65	AC	7.76
172802	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P1-2: LN 556	single	571.52	94.86	100.77	AC	33.73
172920	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	Base Case	single	199.0	98.68	102.81	AC	8.63

9 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172074	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P4-2: 511T556	breaker	699.0	96.17	104.22	AC	56.26
172075	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P4-2: 511T591	breaker	699.0	96.41	104.56	AC	56.26
172023	314691	3FARMVIL	DVP	314692	6FARMVIL	DVP	2	DVP_P4-6: CHASE C T342	breaker	208.7	93.58	135.27	AC	83.05
172091	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 556T591	breaker	684.0	94.63	102.95	AC	56.29
172092	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 511T556	breaker	684.0	94.64	102.97	AC	56.29
172093	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 511T591	breaker	684.0	94.99	103.33	AC	56.29

10 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172751	313867	6BREMODIST	DVP	314765	6MTEAGLE	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	114.93	118.08	AC	20.88
172752	313867	6BREMODIST	DVP	314765	6MTEAGLE	DVP	1	DVP_P1-2: LN 2027-A	single	661.76	110.45	113.6	AC	20.88
171907	314282	6CARSON	DVP	314285	6CHRL249	DVP	1	DVP_P4-2: 562T563	breaker	684.0	129.42	130.51	AC	8.71
171935	314285	6CHRL249	DVP	314316	6LOCKS	DVP	1	DVP_P4-2: 562T563	breaker	684.0	126.64	127.73	AC	8.71
172634	314559	3CAROLNA	DVP	314561	6CAROLNA	DVP	1	DVP_P1-2: LN 1029-A	single	239.89	103.8	105.74	AC	4.6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
171738	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P4-6: CAROLIN T122	breaker	208.6	191.81	198.49	AC	16.75
171739	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P4-2: 2202	breaker	208.6	183.84	190.73	AC	15.79
172463	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-2: LN 130-A	single	182.64	123.92	127.46	AC	8.62
172464	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-3: 6CAROLNA-TX#4	single	182.64	118.41	122.19	AC	9.16
172864	314697	6SEDGE HILL	DVP	927250	AC1-221 TAP	DVP	1	DVP_P1-2: LN 556	single	674.92	104.77	107.9	AC	20.89
171848	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P4-6: CAROLIN T122	breaker	199.0	148.53	156.84	AC	19.9
171849	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P4-2: 2202	breaker	199.0	140.54	147.79	AC	17.39
172918	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P1-2: LN 296-B	single	199.0	108.75	113.31	AC	9.56
172919	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P1-2: LN 570	single	199.0	109.71	113.84	AC	8.66
173346	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	153.59	161.8	AC	19.66
173347	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P7-1: LN 22-90-B	tower	199.0	142.48	150.7	AC	19.66
3518850	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P1-2: LN 570_FSA	single	199.0	109.71	113.84	AC	8.66
3518851	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P1-2: LN 6002_FSA	single	199.0	109.7	113.84	AC	8.66
172708	314747	6BREMO	DVP	313867	6BREMODIST	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	117.21	120.35	AC	20.88
172709	314747	6BREMO	DVP	313867	6BREMODIST	DVP	1	DVP_P1-2: LN 2027-A	single	661.76	112.72	115.87	AC	20.88
172786	314765	6MTEAGLE	DVP	314749	6CHARLV	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	111.79	114.93	AC	20.88
172787	314765	6MTEAGLE	DVP	314749	6CHARLV	DVP	1	DVP_P1-2: LN 2027-A	single	661.76	107.3	110.45	AC	20.88
172765	927250	AC1-221 TAP	DVP	304070	6PERSON230 T	CPL	1	DVP_P1-2: LN 556	single	718.0	108.19	111.08	AC	20.89

11 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
173103	313825	3PLYWOOD	DVP	314696	3SEEDGE HILL	DVP	1	DVP_P1-2: LN 33-B	operation	141.0	108.06	117.09	AC	15.19
172749	313867	6BREMIST	DVP	314765	6MTEAGLE	DVP	1	DVP_P1-2: LN 2027-B	operation	661.76	126.38	131.58	AC	34.8
3518707	314559	3CAROLNA	DVP	314561	6CAROLNA	DVP	1	DVP_P1-3: 6CLUBHSE-TX#1	operation	239.89	158.11	163.82	AC	14.16
172458	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P1-3: 6CAROLNA-TX#4	operation	182.64	183.47	189.87	AC	15.27
172462	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	Base Case	operation	176.81	143.13	149.06	AC	12.94
172801	314677	6BUCKING	DVP	314747	6BREMIO	DVP	1	DVP_P1-2: LN 556	operation	571.52	115.06	124.88	AC	56.22
173042	314687	3CREWE	DVP	314701	3LONEPN	DVP	1	DVP_P1-2: LN 98-A	operation	269.78	65.82	117.58	AC	149.96
172807	314691	3FARMVIL	DVP	314692	6FARMVIL	DVP	2	DVP_P1-2: LN 235-A	operation	182.64	104.87	138.04	AC	62.96
3519153	314691	3FARMVIL	DVP	314692	6FARMVIL	DVP	1	DVP_P1-3: 6FARMVIL-TX#5	operation	198.06	83.6	114.92	AC	65.17
172839	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P1-2: LN 556	operation	559.3	113.06	123.15	AC	56.24
173152	314696	3SEEDGE HILL	DVP	314697	6SEEDGE HILL	DVP	1	DVP_P1-3: 6SEEDGE HILL-TX#2	operation	226.73	99.52	106.81	AC	20.08
172863	314697	6SEEDGE HILL	DVP	927250	AC1-221 TAP	DVP	1	DVP_P1-2: LN 556	operation	674.92	121.63	126.77	AC	34.82
3519154	314701	3LONEPN	DVP	314707	3MORAN	DVP	1	DVP_P1-2: LN 98-A	operation	269.78	60.05	112.79	AC	149.94
172910	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P1-2: LN 296-B	operation	199.0	136.31	142.94	AC	15.93
172914	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	Base Case	operation	199.0	123.51	129.48	AC	14.39
173009	314706	3MARTIN TAP	DVP	314687	3CREWE	DVP	1	DVP_P1-2: LN 98-A	operation	269.78	68.88	120.75	AC	149.96
173214	314707	3MORAN	DVP	314691	3FARMVIL	DVP	1	DVP_P1-2: LN 98-A	operation	203.98	77.31	147.05	AC	149.94
172706	314747	6BREMIO	DVP	313867	6BREMIST	DVP	1	DVP_P1-2: LN 2027-B	operation	661.76	128.67	133.87	AC	34.8
172784	314765	6MTEAGLE	DVP	314749	6CHARVL	DVP	1	DVP_P1-2: LN 2027-B	operation	661.76	121.36	126.56	AC	34.8
172453	314914	8MDLTHAN	DVP	314918	8NO ANNA	DVP	1	Base Case	operation	2442.12	103.79	104.69	AC	25.62
173005	924160	AB2-060 TAP	DVP	314267	3CHASCTY2	DVP	1	DVP_P1-2: LN 1034-A	operation	269.78	69.74	121.29	AC	149.96
172764	927250	AC1-221 TAP	DVP	304070	6PERSON230 T	CPL	1	DVP_P1-2: LN 556	operation	718.0	127.0	131.77	AC	34.82
173111	934610	AD1-087 TAP	DVP	314697	6SEEDGE HILL	DVP	1	DVP_P1-2: LN 556	operation	814.98	99.48	101.52	AC	16.01
173003	939370	AE1-168 TAP	DVP	314706	3MARTIN TAP	DVP	1	DVP_P1-2: LN 98-A	operation	269.78	69.46	121.37	AC	149.96

12 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN																		
172091, 172092, 172093	6	6FARMVIL-6BUCKING 230kV	Description: Rebuild 12.76 miles of the Farmville – Buckingham 230kV line # 298 Final Ratings: 1047/1047/1204 <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$19,140,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-168</td><td>22.47</td><td>100.00%</td><td>\$19,140,000.00</td><td>'DVP_P4-2: 511T591'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost(\$19,140,000)	Contingency Name	Contingency Type	AE1-168	22.47	100.00%	\$19,140,000.00	'DVP_P4-2: 511T591'	breaker	19,140,000	19,140,000	N6245						
Queue	MW contribution	Percentage of Cost	Cost(\$19,140,000)	Contingency Name	Contingency Type																			
AE1-168	22.47	100.00%	\$19,140,000.00	'DVP_P4-2: 511T591'	breaker																			
172023	5	3FARMVIL-6FARMVIL	Description: Add new 230/115 kV transformer to the Farmville substation <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$6,600,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-168</td><td>70.25</td><td>100.00%</td><td>\$6,600,000.00</td><td>DVP_P4-6: CHASE CT342</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost(\$6,600,000)	Contingency Name	Contingency Type	AE1-168	70.25	100.00%	\$6,600,000.00	DVP_P4-6: CHASE CT342	breaker	\$6,600,000	\$6,600,000	N6246						
Queue	MW contribution	Percentage of Cost	Cost(\$6,600,000)	Contingency Name	Contingency Type																			
AE1-168	70.25	100.00%	\$6,600,000.00	DVP_P4-6: CHASE CT342	breaker																			
172786, 172787	12	6MTEAGLE 230.0 kV - 6CHARLVL 230.0 kV Ckt 1	Description : Rebuild 6.47 miles of the Mount Eagle – Charlottesville 230kV line #2028 Time Estimate : 30-36 months Ratings after the upgrade: 1047/1047/1204 <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$16,175,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-056</td><td>10.47</td><td>33.41%</td><td>\$5,403,640.58</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr><tr><td>AE1-168</td><td>20.87</td><td>66.59%</td><td>\$10,771,359.42</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr></table> Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE1-168 could become the driver and could be responsible for the upgrade Note 2: Although Queue Project AE1-168 may not have cost responsibility for this upgrade, Queue Project AE1-168 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-168 comes into service prior to completion of the upgrade, Queue Project AE1-168 will need an interim study.	Queue	MW contribution	Percentage of Cost	Cost (\$16,175,000)	Contingency Name	Contingency Type	AE1-056	10.47	33.41%	\$5,403,640.58	'DVP_P1-2: LN 2027-B'	single	AE1-168	20.87	66.59%	\$10,771,359.42	'DVP_P1-2: LN 2027-B'	single	\$16,175,000	\$10,771,359	N6247
Queue	MW contribution	Percentage of Cost	Cost (\$16,175,000)	Contingency Name	Contingency Type																			
AE1-056	10.47	33.41%	\$5,403,640.58	'DVP_P1-2: LN 2027-B'	single																			
AE1-168	20.87	66.59%	\$10,771,359.42	'DVP_P1-2: LN 2027-B'	single																			

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN																																																												
171907	8	6CARSON-6CHRL249 230kV	Description : Line #249 Carson - Chaparral: wreck and rebuild 7.5 miles of the Carson to Chaparral line #249 to increase its line rating to 1047 MVA (normal), 1047 MVA (emergency), and 1204 MVA (load dump). Time estimate : 44-48 months <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$22,500,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-088</td><td>0.62</td><td>0.42%</td><td>\$93,454.81</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-033</td><td>8.56</td><td>5.73%</td><td>\$1,290,279.36</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>10.63</td><td>7.12%</td><td>\$1,602,297.85</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>11.33</td><td>7.59%</td><td>\$1,707,811.35</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>10.01</td><td>6.71%</td><td>\$1,508,843.04</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-068</td><td>51.71</td><td>34.64%</td><td>\$7,794,432.91</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-069</td><td>41.37</td><td>27.71%</td><td>\$6,235,847.79</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-148</td><td>6.95</td><td>4.66%</td><td>\$1,047,598.31</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-168</td><td>8.71</td><td>5.84%</td><td>\$1,312,889.40</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost(\$22,500,000)	Contingency Name	Contingency Type	AD1-088	0.62	0.42%	\$93,454.81	'DVP_P4-2: 562T563'	breaker	AD2-033	8.56	5.73%	\$1,290,279.36	'DVP_P4-2: 562T563'	breaker	AD2-063	10.63	7.12%	\$1,602,297.85	'DVP_P4-2: 562T563'	breaker	AD2-169	11.33	7.59%	\$1,707,811.35	'DVP_P4-2: 562T563'	breaker	AE1-026	10.01	6.71%	\$1,508,843.04	'DVP_P4-2: 562T563'	breaker	AE1-068	51.71	34.64%	\$7,794,432.91	'DVP_P4-2: 562T563'	breaker	AE1-069	41.37	27.71%	\$6,235,847.79	'DVP_P4-2: 562T563'	breaker	AE1-148	6.95	4.66%	\$1,047,598.31	'DVP_P4-2: 562T563'	breaker	AE1-168	8.71	5.84%	\$1,312,889.40	'DVP_P4-2: 562T563'	breaker	\$22,500,000	\$1,312,889	n6112
Queue	MW contribution	Percentage of Cost	Cost(\$22,500,000)	Contingency Name	Contingency Type																																																													
AD1-088	0.62	0.42%	\$93,454.81	'DVP_P4-2: 562T563'	breaker																																																													
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AE1-168	8.71	5.84%	\$1,312,889.40	'DVP_P4-2: 562T563'	breaker																																																													
134653, 133574	9	6CHRL249-6LOCKS 230kV	Description: Wreck and rebuild 3 miles of the Chaparral – Locks line #249 to increase its line rating to 1047 MVA (normal), 1047 MVA (emergency), and 1204 MVA (load dump). A Virginia CPCN is required. <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$9,000,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD2-033</td><td>5.79</td><td>3.82%</td><td>\$344,242.78</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>10.63</td><td>7.03%</td><td>\$632,549.84</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>11.33</td><td>7.49%</td><td>\$674,204.11</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-026</td><td>14.76</td><td>9.76%</td><td>\$878,310.03</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-068</td><td>51.71</td><td>34.19%</td><td>\$3,077,060.40</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-069</td><td>41.37</td><td>27.35%</td><td>\$2,461,767.33</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-148</td><td>6.95</td><td>4.60%</td><td>\$413,567.39</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr><tr><td>AE1-168</td><td>8.71</td><td>5.76%</td><td>\$518,298.13</td><td>'DVP_P4-2: 562T563'</td><td>breaker</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost(\$9,000,000)	Contingency Name	Contingency Type	AD2-033	5.79	3.82%	\$344,242.78	'DVP_P4-2: 562T563'	breaker	AD2-063	10.63	7.03%	\$632,549.84	'DVP_P4-2: 562T563'	breaker	AD2-169	11.33	7.49%	\$674,204.11	'DVP_P4-2: 562T563'	breaker	AE1-026	14.76	9.76%	\$878,310.03	'DVP_P4-2: 562T563'	breaker	AE1-068	51.71	34.19%	\$3,077,060.40	'DVP_P4-2: 562T563'	breaker	AE1-069	41.37	27.35%	\$2,461,767.33	'DVP_P4-2: 562T563'	breaker	AE1-148	6.95	4.60%	\$413,567.39	'DVP_P4-2: 562T563'	breaker	AE1-168	8.71	5.76%	\$518,298.13	'DVP_P4-2: 562T563'	breaker	\$9,000,000	\$518,298	N6116						
Queue	MW contribution	Percentage of Cost	Cost(\$9,000,000)	Contingency Name	Contingency Type																																																													
AD2-033	5.79	3.82%	\$344,242.78	'DVP_P4-2: 562T563'	breaker																																																													
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ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN																																																												
3518714, 172634	1	3CAROLNA-6CAROLNA 230/115kV	Description: Add a second 230-115 kV transformer at Carolina substation with ratings of 276.8 MVA (normal), 292.4 MVA (emergency), and 328.7 MVA (load dump).	\$5,500,000	\$299,875	N6113																																																												
			<table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$5,500,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-047</td><td>35.10</td><td>22.54%</td><td>\$1,239,906.24</td><td>'DVP_P4-2: 14852'</td><td>breaker</td></tr><tr><td>AD1-050</td><td>9.5</td><td>6.10%</td><td>\$335,548.92</td><td>'DVP_P4-2: 7152'</td><td>breaker</td></tr><tr><td>AD2-033</td><td>11.18</td><td>7.18%</td><td>\$394,888.10</td><td>'DVP_P4-2: 7152'</td><td>breaker</td></tr><tr><td>AD2-046</td><td>12.53</td><td>8.05%</td><td>\$442,571.36</td><td>'DVP_P4-2: 7152'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>14.4</td><td>9.25%</td><td>\$508,621.52</td><td>'DVP_P4-2: 7152'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>44.811</td><td>28.78%</td><td>\$1,582,766.59</td><td>'DVP_P4-2: 7152'</td><td>breaker</td></tr><tr><td>AE1-103</td><td>5.72</td><td>3.67%</td><td>\$202,035.77</td><td>DVP_P4-2: 102952'</td><td>breaker</td></tr><tr><td>AE1-148</td><td>13.98</td><td>8.98%</td><td>\$493,786.73</td><td>DVP_P4-2: 102952'</td><td>breaker</td></tr><tr><td>AE1-168</td><td>8.49</td><td>5.45%</td><td>\$299,874.77</td><td>'DVP_P1-3: 6CLUBHSE-TX#1'</td><td>single</td></tr></table>				Queue	MW contribution	Percentage of Cost	Cost(\$5,500,000)	Contingency Name	Contingency Type	AD1-047	35.10	22.54%	\$1,239,906.24	'DVP_P4-2: 14852'	breaker	AD1-050	9.5	6.10%	\$335,548.92	'DVP_P4-2: 7152'	breaker	AD2-033	11.18	7.18%	\$394,888.10	'DVP_P4-2: 7152'	breaker	AD2-046	12.53	8.05%	\$442,571.36	'DVP_P4-2: 7152'	breaker	AD2-063	14.4	9.25%	\$508,621.52	'DVP_P4-2: 7152'	breaker	AD2-169	44.811	28.78%	\$1,582,766.59	'DVP_P4-2: 7152'	breaker	AE1-103	5.72	3.67%	\$202,035.77	DVP_P4-2: 102952'	breaker	AE1-148	13.98	8.98%	\$493,786.73	DVP_P4-2: 102952'	breaker	AE1-168	8.49	5.45%	\$299,874.77	'DVP_P1-3: 6CLUBHSE-TX#1'	single
			Queue				MW contribution	Percentage of Cost	Cost(\$5,500,000)	Contingency Name	Contingency Type																																																							
			AD1-047				35.10	22.54%	\$1,239,906.24	'DVP_P4-2: 14852'	breaker																																																							
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			AE1-148				13.98	8.98%	\$493,786.73	DVP_P4-2: 102952'	breaker																																																							
			AE1-168				8.49	5.45%	\$299,874.77	'DVP_P1-3: 6CLUBHSE-TX#1'	single																																																							
			171738, 171739, 172463, 172464, 3518602				2	3CLUBHSE-6CLUBHSE 230/115kV	Description: Add a second 230-115 kV transformer at Clubhouse to increase its rating to 276.8 MVA (normal), 292.4 MVA (emergency), and 328.7 MVA (load dump).	\$6,000,000	\$448,654	N6114																																																						
<table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$6.0M)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD1-047</td><td>56.11</td><td>25.06%</td><td>\$1,503,823.21</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD1-050</td><td>5.48</td><td>2.45%</td><td>\$146,871.35</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-033</td><td>17.94</td><td>8.01%</td><td>\$480,816.05</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-046</td><td>10.28</td><td>4.59%</td><td>\$275,517.78</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>23.14</td><td>10.34%</td><td>\$620,183.02</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-169</td><td>79.397</td><td>35.47%</td><td>\$2,127,946.03</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AE1-056</td><td>3.0124</td><td>1.35%</td><td>\$80,736.36</td><td>DVP_P1-2: LN 130-A</td><td>single</td></tr><tr><td>AE1-148</td><td>11.77</td><td>5.26%</td><td>\$315,451.78</td><td>DVP_P4-6: CAROLIN T122</td><td>breaker</td></tr><tr><td>AE1-168</td><td>16.74</td><td>7.48%</td><td>\$448,654.44</td><td>DVP_P4-6: CAROLIN T122</td><td>breaker</td></tr></table>	Queue	MW contribution		Percentage of Cost	Cost(\$6.0M)	Contingency Name			Contingency Type				AD1-047	56.11	25.06%	\$1,503,823.21	'DVP_P4-5: T122C'	breaker	AD1-050	5.48	2.45%	\$146,871.35	'DVP_P4-5: T122C'	breaker	AD2-033	17.94	8.01%	\$480,816.05	'DVP_P4-5: T122C'	breaker	AD2-046	10.28	4.59%	\$275,517.78	'DVP_P4-5: T122C'	breaker	AD2-063	23.14	10.34%	\$620,183.02	'DVP_P4-5: T122C'	breaker	AD2-169	79.397	35.47%	\$2,127,946.03	'DVP_P4-5: T122C'	breaker	AE1-056	3.0124	1.35%	\$80,736.36	DVP_P1-2: LN 130-A	single	AE1-148	11.77	5.26%	\$315,451.78	DVP_P4-6: CAROLIN T122	breaker	AE1-168	16.74	7.48%	\$448,654.44	DVP_P4-6: CAROLIN T122	breaker
Queue	MW contribution	Percentage of Cost		Cost(\$6.0M)	Contingency Name	Contingency Type																																																												
AD1-047	56.11	25.06%		\$1,503,823.21	'DVP_P4-5: T122C'	breaker																																																												
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AD2-046	10.28	4.59%		\$275,517.78	'DVP_P4-5: T122C'	breaker																																																												
AD2-063	23.14	10.34%		\$620,183.02	'DVP_P4-5: T122C'	breaker																																																												
AD2-169	79.397	35.47%		\$2,127,946.03	'DVP_P4-5: T122C'	breaker																																																												
AE1-056	3.0124	1.35%		\$80,736.36	DVP_P1-2: LN 130-A	single																																																												
AE1-148	11.77	5.26%		\$315,451.78	DVP_P4-6: CAROLIN T122	breaker																																																												
AE1-168	16.74	7.48%		\$448,654.44	DVP_P4-6: CAROLIN T122	breaker																																																												
172802, 172074, 172075	3	6BUCKING-6BREMO 230kV		Description: Rebuild 15.51 miles of the Buckingham – Brema 230kV line #298 1047,1047,1204	\$23,265,000	\$23,265,000			N6219																																																									
			<table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$23,265,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-168</td><td>31.47</td><td>100.00%</td><td>\$23,265,000.00</td><td>'DVP_P4-2: 511T591'</td><td>breaker</td></tr></table>	Queue			MW contribution	Percentage of Cost		Cost(\$23,265,000)	Contingency Name	Contingency Type	AE1-168	31.47	100.00%	\$23,265,000.00	'DVP_P4-2: 511T591'	breaker																																																
			Queue	MW contribution			Percentage of Cost	Cost(\$23,265,000)		Contingency Name	Contingency Type																																																							
			AE1-168	31.47			100.00%	\$23,265,000.00		'DVP_P4-2: 511T591'	breaker																																																							

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN																																										
172920, 171848, 171849, 172918, 172919, 173346, 173347, 3518850, 3518851	4	3KERR- 3GW KING TAP 115kV	Dominion Upgrade Description: Line #45 GW King Tap – Kerr Dam 115 kV: This line is a joint tie line between the Dominion Energy and Duke/Progress transmission systems <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$6.123M)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AD2-033</td><td>21.10</td><td>16.33%</td><td>\$999,940</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-046</td><td>26.16</td><td>20.25%</td><td>\$1,239,871</td><td>'DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AD2-063</td><td>27.1455</td><td>21.01%</td><td>\$1,286,579</td><td>DVP_P4-5: T122C'</td><td>breaker</td></tr><tr><td>AE1-056</td><td>6.4038</td><td>4.96%</td><td>\$303,512</td><td>DVP_P7-1: LN 22-90-A</td><td>tower</td></tr><tr><td>AE1-148</td><td>28.72</td><td>22.23%</td><td>\$1,361,204</td><td>DVP_P7-1: LN 22-90-A</td><td>tower</td></tr><tr><td>AE1-168</td><td>19.662</td><td>15.22%</td><td>\$931,894</td><td>DVP_P7-1: LN 22-90-A</td><td>tower</td></tr></table> Duke Upgrade Description: Reconductor 20 miles pf the GW King Tap – Kerr Dam 115kV line and replace terminal equipment. An affected systems study will need to be completed with Duke/Progress to determine impacts what needs to be done by both parties to increase the rating of this transmission line. Time estimate : 48 months Cost : \$40,000,000	Queue	MW contribution	Percentage of Cost	Cost(\$6.123M)	Contingency Name	Contingency Type	AD2-033	21.10	16.33%	\$999,940	'DVP_P4-5: T122C'	breaker	AD2-046	26.16	20.25%	\$1,239,871	'DVP_P4-5: T122C'	breaker	AD2-063	27.1455	21.01%	\$1,286,579	DVP_P4-5: T122C'	breaker	AE1-056	6.4038	4.96%	\$303,512	DVP_P7-1: LN 22-90-A	tower	AE1-148	28.72	22.23%	\$1,361,204	DVP_P7-1: LN 22-90-A	tower	AE1-168	19.662	15.22%	\$931,894	DVP_P7-1: LN 22-90-A	tower	\$6,123,000	\$931,894	N6115
Queue	MW contribution	Percentage of Cost	Cost(\$6.123M)	Contingency Name	Contingency Type																																											
AD2-033	21.10	16.33%	\$999,940	'DVP_P4-5: T122C'	breaker																																											
AD2-046	26.16	20.25%	\$1,239,871	'DVP_P4-5: T122C'	breaker																																											
AD2-063	27.1455	21.01%	\$1,286,579	DVP_P4-5: T122C'	breaker																																											
AE1-056	6.4038	4.96%	\$303,512	DVP_P7-1: LN 22-90-A	tower																																											
AE1-148	28.72	22.23%	\$1,361,204	DVP_P7-1: LN 22-90-A	tower																																											
AE1-168	19.662	15.22%	\$931,894	DVP_P7-1: LN 22-90-A	tower																																											
172751, 172752	7	6BREMODIST- 6MTEAGLE 230kV	Description: Rebuild 18.94 miles of the Brema – Mount Eagle 230kV line #2028 Time estimate : 36-40 months Ratings after the upgrade: 1047/1047/1204 <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost(\$47,350,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-056</td><td>10.4706</td><td>33.40%</td><td>\$15,814,144.23</td><td>DVP_P1-2: LN 2027-B'</td><td>single</td></tr><tr><td>AE1-168</td><td>20.88</td><td>66.60%</td><td>\$31,535,855.77</td><td>DVP_P1-2: LN 2027-B</td><td>single</td></tr></table>	Queue	MW contribution	Percentage of Cost	Cost(\$47,350,000)	Contingency Name	Contingency Type	AE1-056	10.4706	33.40%	\$15,814,144.23	DVP_P1-2: LN 2027-B'	single	AE1-168	20.88	66.60%	\$31,535,855.77	DVP_P1-2: LN 2027-B	single	\$47,350,000	\$31,535,856	N6185																								
Queue	MW contribution	Percentage of Cost	Cost(\$47,350,000)	Contingency Name	Contingency Type																																											
AE1-056	10.4706	33.40%	\$15,814,144.23	DVP_P1-2: LN 2027-B'	single																																											
AE1-168	20.88	66.60%	\$31,535,855.77	DVP_P1-2: LN 2027-B	single																																											

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN																														
172708, 172709	11	6BREMO 230.0 kV - 6BREMODIST 230.0 kV Ckt 1	Description: Rebuild .16 miles Brema – Brema Distribution 230kV line #2028 Time estimate : 30-36 months Ratings after the upgrade: 1047/1047/1204 The potential cost responsibility is as follows: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>Cost (\$500,000)</th><th>Contingency Name</th><th>Contingency Type</th></tr><tr><td>AE1-056</td><td>10.4706</td><td>12.85%</td><td>\$64,255</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr><tr><td>AE1-108</td><td>38.0441</td><td>46.69%</td><td>\$233,467</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr><tr><td>AE1-168</td><td>20.8818</td><td>25.63%</td><td>\$128,146</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr><tr><td>AE1-220</td><td>12.08</td><td>14.83%</td><td>\$74,132</td><td>'DVP_P1-2: LN 2027-B'</td><td>single</td></tr></table> Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE1-168 could become the driver and could be responsible for the upgrade. Note 2: Although Queue Project AE1-168 may not have cost responsibility for this upgrade, Queue Project AE1-168 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-168 comes into service prior to completion of the upgrade, Queue Project AE1-168 will need an interim study.	Queue	MW contribution	Percentage of Cost	Cost (\$500,000)	Contingency Name	Contingency Type	AE1-056	10.4706	12.85%	\$64,255	'DVP_P1-2: LN 2027-B'	single	AE1-108	38.0441	46.69%	\$233,467	'DVP_P1-2: LN 2027-B'	single	AE1-168	20.8818	25.63%	\$128,146	'DVP_P1-2: LN 2027-B'	single	AE1-220	12.08	14.83%	\$74,132	'DVP_P1-2: LN 2027-B'	single	\$500,000	\$0	N5873
Queue	MW contribution	Percentage of Cost	Cost (\$500,000)	Contingency Name	Contingency Type																															
AE1-056	10.4706	12.85%	\$64,255	'DVP_P1-2: LN 2027-B'	single																															
AE1-108	38.0441	46.69%	\$233,467	'DVP_P1-2: LN 2027-B'	single																															
AE1-168	20.8818	25.63%	\$128,146	'DVP_P1-2: LN 2027-B'	single																															
AE1-220	12.08	14.83%	\$74,132	'DVP_P1-2: LN 2027-B'	single																															
172864	10	6SEDGE HILL- AC1-221 TAP 230kV	Description : Rebuild 2 miles of the AC1-221 Tap – Sedge Hill 230kV line #33 Time estimate : 30-36 months Ratings after the upgrade: 1047/1047/1204 Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE1-168 could become the driver and could be responsible for the upgrade Note 2: Although Queue Project AE1-168 may not have cost responsibility for this upgrade, Queue Project AE1-168 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-168 comes into service prior to completion of the upgrade, Queue Project AE1-168 will need an interim study.	\$3,000,000	\$0	N6216																														

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AE1-168	NUN
172765	13	AC1-221 TAP-6PERSON230 T 230kV	Dominion Scope Description: Rebuild the Dominion portion (2 miles) of the AC1-221 Tap – Person 230kV line #33 Cost : \$ 3,000,000 Time estimate : 30-36 months Ratings after the upgrade: 1047/1047/1204 Duke Scope Reconductor 4.85 miles of the AC1-221 Tap – Person 230kV line #33 and replace terminal equipment at Person substation. An affected systems study will need to be completed with Duke/Progress to determine impacts and what needs to be done by both parties to increase the rating of this transmission line. Cost : \$ 21,000,000 Time Estimate : 30-36 months Ratings after the upgrade: 1195/1195/1195 Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., Queue Project AE1-168 could become the driver and could be responsible for the upgrade Note 2: Although Queue Project AE1-168 may not have cost responsibility for this upgrade, Queue Project AE1-168 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE1-168 comes into service prior to completion of the upgrade, Queue Project AE1-168 will need an interim study.	\$24,000,000	\$0	N6215
			TOTAL COST	\$189,153,000	\$94,823,825	

13 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

13.1 Contingencies

Contingency Name	Contingency Definition
DVP_P1-2: LN 1029-A	CONTINGENCY 'DVP_P1-2: LN 1029-A' OPEN BRANCH FROM BUS 314562 TO BUS 314570 CKT 1 /* 3CLUBHSE 115.00 - 3METCATP 115.00 OPEN BRANCH FROM BUS 314570 TO BUS 314572 CKT 1 /* 3METCATP 115.00 - 3EMPORIA 115.00 OPEN BRANCH FROM BUS 314570 TO BUS 314588 CKT 1 /* 3METCATP 115.00 - 3METCALF 115.00 OPEN BRANCH FROM BUS 314572 TO BUS 925170 CKT 1 /* 3EMPORIA 115.00 - AB2-174 TAP 115.00 OPEN BRANCH FROM BUS 314572 TO BUS 314863 CKT 1 /* 3EMPORIA 115.00 - 3EMPOR_1 115.00 OPEN BUS 314570 /* ISLAND: 3METCATP 115.00 OPEN BUS 314572 /* ISLAND: 3EMPORIA 115.00 OPEN BUS 314588 /* ISLAND: 3METCALF 115.00 OPEN BUS 314863 /* ISLAND: 3EMPOR_1 115.00 END
DVP_P1-2: LN 98-A	CONTINGENCY 'DVP_P1-2: LN 98-A' OPEN BRANCH FROM BUS 314267 TO BUS 924160 CKT 1 /* 3CHASCTY2 115.00 - AB2-060 TAP 115.00 END
DVP_P7-1: LN 22-90-A	CONTINGENCY 'DVP_P7-1: LN 22-90-A' OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END

Contingency Name	Contingency Definition
DVP_P4-2: 562T563	CONTINGENCY 'DVP_P4-2: 562T563' /* CARSON 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314923 CKT 1 /* 8CARSON 500.00 - 8SEPTA 500.00 OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 END
DVP_P1-2: LN 1034-A	CONTINGENCY 'DVP_P1-2: LN 1034-A' OPEN BRANCH FROM BUS 314432 TO BUS 314706 CKT 1 /* 3MARTIN DP 115.00 - 3MARTIN TAP 115.00 OPEN BRANCH FROM BUS 314687 TO BUS 314701 CKT 1 /* 3CREWE 115.00 - 3LONEPN 115.00 OPEN BRANCH FROM BUS 314687 TO BUS 314706 CKT 1 /* 3CREWE 115.00 - 3MARTIN TAP 115.00 OPEN BRANCH FROM BUS 314706 TO BUS 939370 CKT 1 /* 3MARTIN TAP 115.00 - AE1-168 TAP 115.00 OPEN BUS 314432 /* ISLAND: 3MARTIN DP 115.00 OPEN BUS 314687 /* ISLAND: 3CREWE 115.00 OPEN BUS 314706 /* ISLAND: 3MARTIN TAP 115.00 END
DVP_P1-3: 6CAROLNA-TX#4	CONTINGENCY 'DVP_P1-3: 6CAROLNA-TX#4' OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P1-2: LN 130-A	CONTINGENCY 'DVP_P1-2: LN 130-A' OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314595 TO BUS 314600 CKT 1 /* 3PL HILL 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314600 TO BUS 937570 CKT 1 /* 3PLHITP 115.00 - AD2-169 TAP 115.00 OPEN BUS 314595 /* ISLAND: 3PL HILL 115.00 OPEN BUS 314600 /* ISLAND: 3PLHITP 115.00 END
DVP_P1-2: LN 33-B	CONTINGENCY 'DVP_P1-2: LN 33-B' OPEN BRANCH FROM BUS 926270 TO BUS 314696 CKT 1 /* AC1-105 TAP 115.00 - 3SEEDGE HILL 115.00 END
DVP_P1-2: LN 296-B	CONTINGENCY 'DVP_P1-2: LN 296-B' OPEN BRANCH FROM BUS 927250 TO BUS 314697 CKT 1 /* AC1-221 TAP T230.00 - 6SEEDGE HILL 230.00 END
DVP_P1-2: LN 570_FSA	CONTINGENCY 'DVP_P1-2: LN 570_FSA' OPEN BRANCH FROM BUS 304183 TO BUS 918500 CKT 1 /* 8WAKE 500 TT500.00 - AA1-064 TAP 500.00 END

Contingency Name	Contingency Definition
DVP_P1-2: LN 2027-A	CONTINGENCY 'DVP_P1-2: LN 2027-A' OPEN BRANCH FROM BUS 313868 TO BUS 933500 CKT 1 /* 6CARTERV 230.00 - AC2-165 TAP 230.00 OPEN BRANCH FROM BUS 313868 TO BUS 314747 CKT 1 /* 6CARTERV 230.00 - 6BREMO 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 313868 /* ISLAND: 6CARTERV 230.00 END
DVP_P1-2: LN 2027-B	CONTINGENCY 'DVP_P1-2: LN 2027-B' OPEN BRANCH FROM BUS 933500 TO BUS 314333 CKT 1 /* AC2-165 TAP 230.00 - 6POWHATN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314322 CKT 1 /* 6JUDES F 230.00 - 6MDLTHAN 230.00 OPEN BRANCH FROM BUS 314310 TO BUS 314333 CKT 1 /* 6JUDES F 230.00 - 6POWHATN 230.00 OPEN BUS 314310 /* ISLAND: 6JUDES F 230.00 OPEN BUS 314333 /* ISLAND: 6POWHATN 230.00 END
DVP_P1-3: 6SEDGE HILL-TX#2	CONTINGENCY 'DVP_P1-3: 6SEDGE HILL-TX#2' OPEN BRANCH FROM BUS 314696 TO BUS 314697 CKT 2 /* 3SEDGE HILL 115.00 - 6SEDGE HILL 230.00 END
DVP_P1-2: LN 6002_FSA	CONTINGENCY 'DVP_P1-2: LN 6002_FSA' OPEN BRANCH FROM BUS 314935 TO BUS 918500 CKT 1 /* 8HERITAGE 500.00 - AA1-064 TAP 500.00 END
DVP_P4-2: 2202	CONTINGENCY 'DVP_P4-2: 2202' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 314571 TO BUS 925780 CKT 1 /* 3EATON F 115.00 - AC1-054 TAP 115.00 /* CONTINGENCY LINE ADDED FOR AE1 BUILD OPEN BUS 314571 /* ISLAND: 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 END
DVP_P4-2: 511T591	CONTINGENCY 'DVP_P4-2: 511T591' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314936 CKT 1 /* 8CARSON 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END

Contingency Name	Contingency Definition
DVP_P4-6: CAROLIN T122	CONTINGENCY 'DVP_P4-6: CAROLIN T122' /* CAROLINA 115 KV OPEN BRANCH FROM BUS 314559 TO BUS 315126 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP2 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 315128 CKT 1 /* 3CAROLNA 115.00 - 1ROARAP4 14.400 OPEN BUS 315126 /* ISLAND: 1ROARAP2 14.400 OPEN BUS 315128 /* ISLAND: 1ROARAP4 14.400 OPEN BRANCH FROM BUS 314559 TO BUS 314571 CKT 1 /* 3CAROLNA 115.00 - 3EATON F 115.00 OPEN BRANCH FROM BUS 313722 TO BUS 314559 CKT 1 /* 3OCCONEECHEE115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314259 TO BUS 314559 CKT Z1 /* 3CAROL56_1 115.00 - 3CAROLNA 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1 /* 3CAROLNA 115.00 - 3HORNRTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314600 CKT 1 /* 3CAROLNA 115.00 - 3PLHITP 115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 /* 3CAROLNA 115.00 - 6CAROLNA 230.00 OPEN BUS 314559 /* 3CAROLNA 115.00 KV END
DVP_P1-2: LN 556	CONTINGENCY 'DVP_P1-2: LN 556' OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 1 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 2 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 3 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314906 TO BUS 314936 CKT 1 /* 8CLOVER 500.00 - 8RAWLINGS 500.00 OPEN BUS 314906 /* ISLAND: 8CLOVER 500.00 OPEN BUS 314915 /* ISLAND: 8CLOVER_STC 500.00 END
DVP_P1-2: LN 570	CONTINGENCY 'DVP_P1-2: LN 570' OPEN BRANCH FROM BUS 304183 TO BUS 918500 CKT 1 /* 8WAKE 500 TT500.00 - AA1-064 TAP 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 918500 CKT 1 /* 8HERITAGE 500.00 - AA1-064 TAP 500.00 OPEN BUS 918500 /* ISLAND: AA1-064 TAP 500.00 END
DVP_P4-2: 556T591	CONTINGENCY 'DVP_P4-2: 556T591' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 1 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 2 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 3 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314906 TO BUS 314936 CKT 1 /* 8CLOVER 500.00 - 8RAWLINGS 500.00 OPEN BUS 314906 /* ISLAND: 8CLOVER 500.00 OPEN BUS 314915 /* ISLAND: 8CLOVER_STC 500.00 OPEN BRANCH FROM BUS 314935 TO BUS 314936 CKT 1 /* 8HERITAGE 500.00 - 8RAWLINGS 500.00 END

Contingency Name	Contingency Definition
DVP_P7-1: LN 22-90-B	CONTINGENCY 'DVP_P7-1: LN 22-90-B' OPEN BRANCH FROM BUS 925780 TO BUS 314702 CKT 1 /* AC1-054 TAP 115.00 - 3KERR 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314584 CKT 1 /* 3FIVEFORKSDP115.00 - 3LITTLTN 115.00 OPEN BRANCH FROM BUS 314265 TO BUS 314673 CKT 1 /* 3FIVEFORKSDP115.00 - 3PALMERSPRNG115.00 OPEN BRANCH FROM BUS 314559 TO BUS 314585 CKT 1 /* 3CAROLNA 115.00 - 3L GASTN 115.00 OPEN BRANCH FROM BUS 314584 TO BUS 314585 CKT 1 /* 3LITTLTN 115.00 - 3L GASTN 115.00 OPEN BUS 314265 /* ISLAND: 3FIVEFORKSDP115.00 OPEN BUS 314584 /* ISLAND: 3LITTLTN 115.00 OPEN BUS 314585 /* ISLAND: 3L GASTN 115.00 END
DVP_P1-3: 6FARMVIL-TX#5	CONTINGENCY 'DVP_P1-3: 6FARMVIL-TX#5' OPEN BRANCH FROM BUS 314691 TO BUS 314692 CKT 2 /* 3FARMVIL 115.00 - 6FARMVIL 230.00 END
DVP_P4-6: CHASE C T342	CONTINGENCY 'DVP_P4-6: CHASE C T342' /* CHASE CITY 115 KV OPEN BUS 314267 /* 3CHASCTY2 115.00 KV OPEN BUS 314518 /* ISLAND: 3CHASC_1 115.00 OPEN BUS 314681 /* 3CHASCTY 115.00 KV END
Base Case	
DVP_P4-2: 511T556	CONTINGENCY 'DVP_P4-2: 511T556' /* RAWLINGS 500 KV OPEN BRANCH FROM BUS 314902 TO BUS 314936 CKT 1 /* 8CARSON 500.00 - 8RAWLINGS 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 1 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 2 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314686 TO BUS 314906 CKT 3 /* 6CLOVER 230.00 - 8CLOVER 500.00 OPEN BRANCH FROM BUS 314906 TO BUS 314936 CKT 1 /* 8CLOVER 500.00 - 8RAWLINGS 500.00 OPEN BUS 314906 /* ISLAND: 8CLOVER 500.00 OPEN BUS 314915 /* ISLAND: 8CLOVER_STC 500.00 END
DVP_P1-2: LN 235-A	CONTINGENCY 'DVP_P1-2: LN 235-A' OPEN BRANCH FROM BUS 313802 TO BUS 314268 CKT 1 /* 6PRINCE EDW 230.00 - 6BRIERY 230.00 OPEN BRANCH FROM BUS 313802 TO BUS 314692 CKT 1 /* 6PRINCE EDW 230.00 - 6FARMVIL 230.00 OPEN BRANCH FROM BUS 314268 TO BUS 934620 CKT 1 /* 6BRIERY 230.00 - AD1-088 TAP 230.00 OPEN BRANCH FROM BUS 314691 TO BUS 314692 CKT 1 /* 3FARMVIL 115.00 - 6FARMVIL 230.00 OPEN BUS 313802 /* ISLAND: 6PRINCE EDW 230.00 OPEN BUS 314268 /* ISLAND: 6BRIERY 230.00 END

Contingency Name	Contingency Definition
DVP_P1-3: 6CLUBHSE-TX#1	CONTINGENCY 'DVP_P1-3: 6CLUBHSE-TX#1' OPEN BRANCH FROM BUS 314562 TO BUS 314563 CKT 1 /* 3CLUBHSE 115.00 - 6CLUBHSE 230.00 END

13.2 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172634	314559	3CAROLNA	DVP	314561	6CAROLNA	DVP	1	DVP_P1-2: LN 1029-A	single	239.89	103.8	105.74	AC	4.6

Bus #	Bus	MW Impact
314589	3MURPHYS	0.08
315115	1S HAMPT1	1.2
315126	1ROARAP2	3.15
315128	1ROARAP4	3.03
315150	1BUGGS 1	6.59
315151	1BUGGS 2	6.59
315158	1KERR 1	0.28
315159	1KERR 2	1.14
315160	1KERR 3	1.12
315161	1KERR 4	1.12
315162	1KERR 5	1.12
315163	1KERR 6	1.12
315164	1KERR 7	1.12
917331	Z2-043 C	0.21
917341	Z2-044 C	0.07
918491	AA1-063AC OP	2.15
918561	AA1-072 C	0.03
919691	AA2-053 C	2.71
919701	AA2-057 C	0.68
920041	AA2-088 C OP	1.16
920591	AA2-165 C	0.09
920671	AA2-174 C	0.12
923801	AB2-015 C O1	7.22
923911	AB2-031 C O1	6.02
923991	AB2-040 C O1	2.79
924021	AB2-043 C O1	0.2
924161	AB2-060 C O1	0.58
924301	AB2-077 C O1	0.13
924311	AB2-078 C O1	0.13
924321	AB2-079 C O1	0.13
924401	AB2-089 C	2.63
924501	AB2-099 C	0.35
925171	AB2-174 C O1	18.86
925611	AC1-036 C	0.05
925781	AC1-054 C O1	10.08
926201	AC1-098 C	5.83
926211	AC1-099 C	1.95
927141	AC1-208 C	11.31
931231	AB1-173 C	0.86
931241	AB1-173AC	0.86
932631	AC2-084 C	8.31

Bus #	Bus	MW Impact
934201	AD1-047 C	21.55
934231	AD1-050 C	5.8
936261	AD2-033 C	4.93
936361	AD2-046 C O1	7.7
936401	AD2-051 C O1	3.91
936481	AD2-063 C O1	6.31
936711	AD2-090 C O1	7.43
937571	AD2-169 C	26.94
938771	AE1-103 C O1	2.43
939181	AE1-148 C O1	7.39
939371	AE1-168 C	4.6
CARR	CARR	0.04
CBM-S1	CBM-S1	0.23
CBM-S2	CBM-S2	0.23
CBM-W1	CBM-W1	0.2
CBM-W2	CBM-W2	1.44
CIN	CIN	0.1
CPLE	CPLE	0.09
IPL	IPL	0.06
LGEE	LGEE	0.03
MEC	MEC	0.22
MECS	MECS	0.06
RENSSELAER	RENSSELAER	0.03
WEC	WEC	0.03

13.3 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
171738	314562	3CLUBHSE	DVP	314563	6CLUBHSE	DVP	1	DVP_P4-6: CAROLIN T122	breaker	208.6	191.81	198.49	AC	16.75

Bus #	Bus	MW Impact
314572	3EMPORIA	1.98
314704	3LAWRENC	1.03
315150	1BUGGS 1	8.31
315151	1BUGGS 2	8.31
315158	1KERR 1	0.22
315159	1KERR 2	0.91
315160	1KERR 3	0.9
315161	1KERR 4	0.9
315162	1KERR 5	0.9
315163	1KERR 6	0.9
315164	1KERR 7	0.9
923911	AB2-031 C O1	10.64
923912	AB2-031 E O1	5.24
923991	AB2-040 C O1	4.94
923992	AB2-040 E O1	28.58
924021	AB2-043 C O1	0.45
924022	AB2-043 E O1	5.17
924161	AB2-060 C O1	1.26
924162	AB2-060 E O1	4.21
924301	AB2-077 C O1	0.27
924302	AB2-077 E O1	1.29
924311	AB2-078 C O1	0.27
924312	AB2-078 E O1	1.29
924321	AB2-079 C O1	0.27
924322	AB2-079 E O1	1.29
924401	AB2-089 C	1.61
924402	AB2-089 E	0.83
925171	AB2-174 C O1	33.34
925172	AB2-174 E O1	30.17
925611	AC1-036 C	0.11
925612	AC1-036 E	1.22
925781	AC1-054 C O1	5.43
925782	AC1-054 E O1	2.5
931231	AB1-173 C	1.51
931232	AB1-173 E	5.0
931241	AB1-173AC	1.51
931242	AB1-173AE	5.0
934201	AD1-047 C	38.11
934202	AD1-047 E	25.4
934231	AD1-050 C	3.55

Bus #	Bus	MW Impact
934232	AD1-050 E	1.94
936261	AD2-033 C	10.77
936262	AD2-033 E	7.18
936361	AD2-046 C O1	7.05
936362	AD2-046 E O1	3.24
936481	AD2-063 C O1	13.78
936482	AD2-063 E O1	9.19
937571	AD2-169 C	47.63
937572	AD2-169 E	31.76
939181	AE1-148 C O1	7.07
939182	AE1-148 E O1	4.71
939371	AE1-168 C	10.05
939372	AE1-168 E	6.7
CARR	CARR	0.04
CBM-S1	CBM-S1	0.53
CBM-S2	CBM-S2	0.43
CBM-W1	CBM-W1	0.63
CBM-W2	CBM-W2	3.51
CIN	CIN	0.29
CPL	CPL	0.17
G-007	G-007	0.13
IPL	IPL	0.19
LGEE	LGEE	0.09
MEC	MEC	0.59
MECS	MECS	0.29
O-066	O-066	0.85
RENSSELAER	RENSSELAER	0.03
WEC	WEC	0.08

13.4 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172075	314677	6BUCKING	DVP	314747	6BREMO	DVP	1	DVP_P4-2: 511T591	breaker	699.0	96.41	104.56	AC	56.26

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.81
315150	1BUGGS 1	13.28
315151	1BUGGS 2	13.28
315153	1CLOVER1	18.48
315154	1CLOVER2	18.24
315159	1KERR 2	1.2
315164	1KERR 7	1.19
315266	1PLYWOOD A	1.24
924021	AB2-043 C O1	0.61
924022	AB2-043 E O1	7.09
924031	AB2-045 C	1.33
924032	AB2-045 E	4.39
924161	AB2-060 C O1	1.8
924162	AB2-060 E O1	6.0
924301	AB2-077 C O1	0.38
924302	AB2-077 E O1	1.8
924311	AB2-078 C O1	0.38
924312	AB2-078 E O1	1.8
924321	AB2-079 C O1	0.38
924322	AB2-079 E O1	1.8
924401	AB2-089 C	1.92
924402	AB2-089 E	0.99
925611	AC1-036 C	0.26
925612	AC1-036 E	3.0
925781	AC1-054 C O1	6.07
925782	AC1-054 E O1	2.8
925831	AC1-062	0.11
925991	AC1-075 C	4.97
925992	AC1-075 E	2.82
926021	AC1-080 C	1.66
926022	AC1-080 E	0.93
926271	AC1-105 C O1	6.78
926272	AC1-105 E O1	3.38
927251	AC1-221 C	1.67
927252	AC1-221 E	1.67
927261	AC1-222 C	4.08
927262	AC1-222 E	3.88
932511	AC2-071 C	1.31
932512	AC2-071 E	4.67
932761	AC2-100 C	3.84
932762	AC2-100 E	1.88

Bus #	Bus	MW Impact
934231	AD1-050 C	4.23
934232	AD1-050 E	2.31
934311	AD1-055 C	2.83
934312	AD1-055 E	0.73
934341	AD1-058 C	4.18
934342	AD1-058 E	1.06
934611	AD1-087 C O1	12.07
934612	AD1-087 E O1	5.67
934621	AD1-088 C	29.91
934622	AD1-088 E	14.04
934991	AD1-131 C	1.37
934992	AD1-131 E	0.91
935171	AD1-152 C O1	12.0
935172	AD1-152 E O1	8.0
935221	AD1-157 C	0.55
935222	AD1-157 E	2.59
935231	AD1-160 C	2.85
935232	AD1-160 E	3.93
936261	AD2-033 C	23.77
936262	AD2-033 E	15.84
936331	AD2-043 C	5.29
936332	AD2-043 E	6.26
936361	AD2-046 C O1	9.36
936362	AD2-046 E O1	4.31
936481	AD2-063 C O1	23.83
936482	AD2-063 E O1	15.89
937481	AD2-202 C O1	3.32
937482	AD2-202 E O1	1.67
938371	AE1-056 C	16.75
938372	AE1-056 E	9.15
939181	AE1-148 C O1	9.41
939182	AE1-148 E O1	6.27
939371	AE1-168 C	33.76
939372	AE1-168 E	22.51
AA2-074	AA2-074	2.39
CARR	CARR	0.25
CBM-S1	CBM-S1	5.41
CBM-S2	CBM-S2	6.67
CBM-W1	CBM-W1	5.03
CBM-W2	CBM-W2	34.83
CIN	CIN	2.3
CPL	CPL	3.51
G-007	G-007	0.84
IPL	IPL	1.43
LGEE	LGEE	0.67
MEC	MEC	5.24
MECS	MECS	1.97
O-066	O-066	5.33
RENSSELAER	RENSSELAER	0.2
WEC	WEC	0.61

13.5 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
173346	314702	3KERR	DVP	304102	3GW KING TAP	CPL	1	DVP_P7-1: LN 22-90-A	tower	199.0	153.59	161.8	AC	19.66

Bus #	Bus	MW Impact
315150	1BUGGS 1	15.54
315151	1BUGGS 2	15.54
315158	1KERR 1	0.64
315159	1KERR 2	2.62
315160	1KERR 3	2.59
315161	1KERR 4	2.59
315162	1KERR 5	2.59
315163	1KERR 6	2.59
315164	1KERR 7	2.59
924022	AB2-043 E O1	6.02
924162	AB2-060 E O1	4.91
924302	AB2-077 E O1	1.58
924312	AB2-078 E O1	1.58
924322	AB2-079 E O1	1.58
924401	AB2-089 C	4.63
924402	AB2-089 E	2.38
925612	AC1-036 E	1.43
925781	AC1-054 C O1	15.6
925782	AC1-054 E O1	7.18
926271	AC1-105 C O1	2.87
926272	AC1-105 E O1	1.43
934231	AD1-050 C	10.2
934232	AD1-050 E	5.57
935222	AD1-157 E	0.64
935231	AD1-160 C	0.7
935232	AD1-160 E	0.97
936261	AD2-033 C	12.59
936262	AD2-033 E	8.39
936361	AD2-046 C O1	17.92
936362	AD2-046 E O1	8.24
936481	AD2-063 C O1	16.07
936482	AD2-063 E O1	10.72
938371	AE1-056 C	4.14
938372	AE1-056 E	2.26
939181	AE1-148 C O1	17.24
939182	AE1-148 E O1	11.49
939371	AE1-168 C	11.8
939372	AE1-168 E	7.86
BLUEG	BLUEG	1.98
CALDERWOOD	CALDERWOOD	0.42
CANNELTON	CANNELTON	0.14

Bus #	Bus	MW Impact
CATAWBA	CATAWBA	0.43
CBM-N	CBM-N	0.0
CHEOAH	CHEOAH	0.39
CHILHOWEE	CHILHOWEE	0.14
COFFEEN	COFFEEN	0.23
COTTONWOOD	COTTONWOOD	1.39
DUCKCREEK	DUCKCREEK	0.48
EDWARDS	EDWARDS	0.21
FARMERCITY	FARMERCITY	0.17
G-007A	G-007A	0.09
GIBSON	GIBSON	0.09
HAMLET	HAMLET	0.81
NEWTON	NEWTON	0.61
NYISO	NYISO	0.02
PRAIRIE	PRAIRIE	1.28
SANTEETLA	SANTEETLA	0.12
SMITHLAND	SMITHLAND	0.11
TATANKA	TATANKA	0.29
TILTON	TILTON	0.25
TRIMBLE	TRIMBLE	0.22
TVA	TVA	1.17
UNIONPOWER	UNIONPOWER	0.59
VFT	VFT	0.22

13.6 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172023	314691	3FARMVIL	DVP	314692	6FARMVIL	DVP	2	DVP_P4-6: CHASE C T342	breaker	208.7	93.58	135.27	AC	83.05

Bus #	Bus	MW Impact
314429	3JTRSVLE	1.1
924161	AB2-060 C O1	4.26
924162	AB2-060 E O1	14.17
925611	AC1-036 C	0.45
925612	AC1-036 E	5.15
925831	AC1-062	0.15
935221	AD1-157 C	0.7
935222	AD1-157 E	3.32
935231	AD1-160 C	3.65
935232	AD1-160 E	5.04
936261	AD2-033 C	43.18
936262	AD2-033 E	28.79
936481	AD2-063 C O1	49.83
936482	AD2-063 E O1	33.22
938371	AE1-056 C	21.48
938372	AE1-056 E	11.74
939371	AE1-168 C	49.83
939372	AE1-168 E	33.22
BLUEG	BLUEG	0.13
CALDERWOOD	CALDERWOOD	0.01
CANNELTON	CANNELTON	0.01
CARR	CARR	0.01
CATAWBA	CATAWBA	0.01
CHEOAH	CHEOAH	0.01
CHILHOWEE	CHILHOWEE	0.0
COFFEEN	COFFEEN	0.01
COTTONWOOD	COTTONWOOD	0.05
DUCKCREEK	DUCKCREEK	0.03
EDWARDS	EDWARDS	0.01
FARMERCITY	FARMERCITY	0.01
G-007	G-007	0.03
GIBSON	GIBSON	0.01
HAMLET	HAMLET	0.01
NEWTON	NEWTON	0.03
O-066	O-066	0.17
PRAIRIE	PRAIRIE	0.06
RENSSELAER	RENSSELAER	0.01
SANTEETLA	SANTEETLA	0.0
SMITHLAND	SMITHLAND	0.01
TATANKA	TATANKA	0.02

Bus #	Bus	MW Impact
TILTON	TILTON	0.02
TRIMBLE	TRIMBLE	0.01
TVA	TVA	0.04
UNIONPOWER	UNIONPOWER	0.02

13.7 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172093	314692	6FARMVIL	DVP	314677	6BUCKING	DVP	1	DVP_P4-2: 511T591	breaker	684.0	94.99	103.33	AC	56.29

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.81
315150	1BUGGS 1	13.29
315151	1BUGGS 2	13.29
315153	1CLOVER1	18.49
315154	1CLOVER2	18.25
315159	1KERR 2	1.2
315164	1KERR 7	1.19
315266	1PLYWOOD A	1.24
924021	AB2-043 C O1	0.61
924022	AB2-043 E O1	7.09
924161	AB2-060 C O1	1.8
924162	AB2-060 E O1	6.0
924301	AB2-077 C O1	0.38
924302	AB2-077 E O1	1.8
924311	AB2-078 C O1	0.38
924312	AB2-078 E O1	1.8
924321	AB2-079 C O1	0.38
924322	AB2-079 E O1	1.8
924401	AB2-089 C	1.92
924402	AB2-089 E	0.99
925611	AC1-036 C	0.26
925612	AC1-036 E	3.0
925781	AC1-054 C O1	6.08
925782	AC1-054 E O1	2.8
925831	AC1-062	0.11
925991	AC1-075 C	4.98
925992	AC1-075 E	2.82
926021	AC1-080 C	1.66
926022	AC1-080 E	0.94
926271	AC1-105 C O1	6.79
926272	AC1-105 E O1	3.38
927251	AC1-221 C	1.67
927252	AC1-221 E	1.67
927261	AC1-222 C	4.08
927262	AC1-222 E	3.89
932761	AC2-100 C	3.85
932762	AC2-100 E	1.88
934231	AD1-050 C	4.23
934232	AD1-050 E	2.31
934311	AD1-055 C	2.83
934312	AD1-055 E	0.73

Bus #	Bus	MW Impact
934341	AD1-058 C	4.19
934342	AD1-058 E	1.06
934611	AD1-087 C O1	12.08
934612	AD1-087 E O1	5.68
934621	AD1-088 C	29.92
934622	AD1-088 E	14.05
934991	AD1-131 C	1.37
934992	AD1-131 E	0.92
935171	AD1-152 C O1	12.0
935172	AD1-152 E O1	8.0
935221	AD1-157 C	0.55
935222	AD1-157 E	2.59
935231	AD1-160 C	2.85
935232	AD1-160 E	3.93
936261	AD2-033 C	23.78
936262	AD2-033 E	15.85
936331	AD2-043 C	5.29
936332	AD2-043 E	6.26
936361	AD2-046 C O1	9.37
936362	AD2-046 E O1	4.31
936481	AD2-063 C O1	23.85
936482	AD2-063 E O1	15.9
937481	AD2-202 C O1	3.33
937482	AD2-202 E O1	1.68
938371	AE1-056 C	16.75
938372	AE1-056 E	9.15
939181	AE1-148 C O1	9.42
939182	AE1-148 E O1	6.28
939371	AE1-168 C	33.77
939372	AE1-168 E	22.51
AA2-074	AA2-074	2.4
CARR	CARR	0.25
CBM-S1	CBM-S1	5.45
CBM-S2	CBM-S2	6.69
CBM-W1	CBM-W1	5.12
CBM-W2	CBM-W2	35.18
CIN	CIN	2.33
CPL	CPL	3.52
G-007	G-007	0.82
IPL	IPL	1.46
LGEE	LGEE	0.68
MEC	MEC	5.31
MECS	MECS	2.02
O-066	O-066	5.23
RENSSELAER	RENSSELAER	0.19
WEC	WEC	0.62

13.8 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172751	313867	6BREMODIST	DVP	314765	6MTEAGLE	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	114.93	118.08	AC	20.88

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.5
314704	3LAWRENC	0.13
315150	1BUGGS 1	7.33
315151	1BUGGS 2	7.33
315153	1CLOVER1	6.11
315154	1CLOVER2	6.03
315158	1KERR 1	0.17
315159	1KERR 2	0.71
315160	1KERR 3	0.7
315161	1KERR 4	0.7
315162	1KERR 5	0.7
315163	1KERR 6	0.7
315164	1KERR 7	0.7
315170	1BREMO 3	30.25
315171	1BREMO 4	66.47
315191	1BEARGRDN G1	12.56
315192	1BEARGRDN G2	12.56
315193	1BEARGRDN S1	25.85
315266	1PLYWOOD A	0.57
920291	AA2-127	2.61
923911	AB2-031 C O1	0.67
923991	AB2-040 C O1	0.31
924021	AB2-043 C O1	0.36
924031	AB2-045 C	0.89
924161	AB2-060 C O1	1.06
924301	AB2-077 C O1	0.22
924311	AB2-078 C O1	0.22
924321	AB2-079 C O1	0.22
924401	AB2-089 C	1.15
925171	AB2-174 C O1	2.12
925611	AC1-036 C	0.16
925781	AC1-054 C O1	3.68
925831	AC1-062	0.07
925991	AC1-075 C	2.53
926021	AC1-080 C	0.85
926271	AC1-105 C O1	3.21
927251	AC1-221 C	0.76
927261	AC1-222 C	1.83
931231	AB1-173 C	0.1
931241	AB1-173AC	0.1
932511	AC2-071 C	0.88

Bus #	Bus	MW Impact
932761	AC2-100 C	1.74
933501	AC2-165 C	32.69
934201	AD1-047 C	2.4
934231	AD1-050 C	2.53
934311	AD1-055 C	1.27
934341	AD1-058 C	1.9
934611	AD1-087 C O1	4.25
934621	AD1-088 C	14.59
934991	AD1-131 C	0.62
935171	AD1-152 C O1	4.23
935221	AD1-157 C	0.34
935231	AD1-160 C	1.78
936261	AD2-033 C	14.42
936331	AD2-043 C	2.38
936361	AD2-046 C O1	5.5
936481	AD2-063 C O1	14.24
937481	AD2-202 C O1	1.17
938371	AE1-056 C	10.47
938821	AE1-108 C O1	38.04
939181	AE1-148 C O1	5.53
939371	AE1-168 C	20.88
939841	AE1-220 C O1	12.08
AA2-074	AA2-074	1.48
CARR	CARR	0.29
CBM-S1	CBM-S1	2.82
CBM-S2	CBM-S2	4.08
CBM-W1	CBM-W1	1.69
CBM-W2	CBM-W2	17.19
CIN	CIN	0.85
CPLE	CPLE	2.17
IPL	IPL	0.51
LGEE	LGEE	0.25
MEC	MEC	2.24
MECS	MECS	0.24
RENSSELAER	RENSSELAER	0.23
WEC	WEC	0.22

13.9 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
171907	314282	6CARSON	DVP	314285	6CHRL249	DVP	1	DVP_P4-2: 562T563	breaker	684.0	129.42	130.51	AC	8.71

Bus #	Bus	MW Impact
314541	3WATKINS	0.23
314554	3BTLEBRO	0.4
314557	3BETHEL	0.39
314572	3EMPORIA	0.33
314574	6EVERETS	1.77
314578	3HORNRTN	1.92
314582	3KELFORD	1.62
314603	3SCOT NK	1.61
314617	3TUNIS	0.36
314620	6CASHIE	0.31
314623	3WITAKRS	0.69
314704	3LAWRENC	0.2
315102	1BRUNSWICKG1	3.87
315103	1BRUNSWICKG2	3.87
315104	1BRUNSWICKG3	3.87
315105	1BRUNSWICKS1	8.03
315131	1EDGECEMA	4.73
315132	1EDGECEMB	4.73
315136	1ROSEMG1	1.22
315137	1ROSEMS1	0.76
315138	1ROSEMG2	0.57
315139	1GASTONA	1.77
315141	1GASTONB	1.77
315150	1BUGGS 1	5.14
315151	1BUGGS 2	5.14
900672	V4-068 E	0.13
916301	Z1-086 C	23.58
916302	Z1-086 E	5.22
917332	Z2-043 E	0.46
917342	Z2-044 E	0.3
917512	Z2-088 E OP1	1.76
918492	AA1-063AE OP	2.11
918512	AA1-065 E OP	1.81
918532	AA1-067 E	0.31
918562	AA1-072 E	0.08
919692	AA2-053 E	2.07
919702	AA2-057 E	1.83
920042	AA2-088 E OP	4.27
920592	AA2-165 E	0.24
920672	AA2-174 E	0.24
923851	AB2-025 C	0.41

Bus #	Bus	MW Impact
923852	AB2-025 E	1.3
923911	AB2-031 C O1	1.6
923912	AB2-031 E O1	0.79
923991	AB2-040 C O1	0.74
923992	AB2-040 E O1	4.29
924022	AB2-043 E O1	2.32
924152	AB2-059 E O1	2.75
924162	AB2-060 E O1	1.9
924302	AB2-077 E O1	0.6
924312	AB2-078 E O1	0.6
924322	AB2-079 E O1	0.6
924401	AB2-089 C	1.07
924402	AB2-089 E	0.55
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.25
924502	AB2-099 E	0.11
924511	AB2-100 C	10.62
924512	AB2-100 E	5.23
925122	AB2-169 E	2.2
925171	AB2-174 C O1	5.2
925172	AB2-174 E O1	4.7
925591	AC1-034 C	3.46
925592	AC1-034 E	2.61
925612	AC1-036 E	0.6
925781	AC1-054 C O1	3.69
925782	AC1-054 E O1	1.7
926071	AC1-086 C	14.05
926072	AC1-086 E	6.39
926201	AC1-098 C	3.11
926202	AC1-098 E	1.86
926211	AC1-099 C	1.04
926212	AC1-099 E	0.61
926271	AC1-105 C O1	2.36
926272	AC1-105 E O1	1.18
927021	AC1-189 C	3.85
927022	AC1-189 E	1.92
927141	AC1-208 C	4.79
927142	AC1-208 E	2.13
927251	AC1-221 C	0.93
927252	AC1-221 E	0.93
927261	AC1-222 C	1.47
927262	AC1-222 E	1.4
930402	AB1-081 E O1	1.94
930861	AB1-132 C O1	1.35
930862	AB1-132 E O1	4.09
931231	AB1-173 C	0.23
931232	AB1-173 E	0.75
931241	AB1-173AC	0.23
931242	AB1-173AE	0.75
932631	AC2-084 C	4.44
932632	AC2-084 E	2.19

Bus #	Bus	MW Impact
932761	AC2-100 C	2.13
932762	AC2-100 E	1.04
933991	AD1-023 C	4.64
933992	AD1-023 E	2.53
934201	AD1-047 C	5.72
934202	AD1-047 E	3.81
934231	AD1-050 C	2.36
934232	AD1-050 E	1.29
934311	AD1-055 C	1.02
934312	AD1-055 E	0.26
934331	AD1-057 C O1	5.86
934332	AD1-057 E O1	3.12
934341	AD1-058 C	2.33
934342	AD1-058 E	0.59
934521	AD1-076 C	18.15
934522	AD1-076 E	9.24
934611	AD1-087 C O1	3.67
934612	AD1-087 E O1	1.72
934621	AD1-088 C	4.84
934622	AD1-088 E	2.27
934991	AD1-131 C	0.76
934992	AD1-131 E	0.51
935171	AD1-152 C O1	3.65
935172	AD1-152 E O1	2.43
935212	AD1-156 E	0.63
936261	AD2-033 C	5.16
936262	AD2-033 E	3.44
936331	AD2-043 C	1.91
936332	AD2-043 E	2.26
936361	AD2-046 C O1	4.24
936362	AD2-046 E O1	1.95
936401	AD2-051 C O1	3.84
936402	AD2-051 E O1	1.65
936481	AD2-063 C O1	6.36
936482	AD2-063 E O1	4.24
936531	AD2-068 C	2.27
936532	AD2-068 E	1.17
936701	AD2-089 C	3.47
936702	AD2-089 E	2.31
936711	AD2-090 C O1	2.58
936712	AD2-090 E O1	1.72
937481	AD2-202 C O1	1.01
937482	AD2-202 E O1	0.51
937571	AD2-169 C	6.85
937572	AD2-169 E	4.56
938171	AE1-026 C1 O	10.2
938172	AE1-026 C2 O	1.57
938173	AE1-026 E O1	2.99
938221	AE1-035 C	0.92
938222	AE1-035 E	0.45
938491	AE1-068 C O1	33.31
938492	AE1-068 E O1	18.4

Bus #	Bus	MW Impact
938501	AE1-069 C O1	26.32
938502	AE1-069 E O1	15.05
938661	AE1-088	1.26
939181	AE1-148 C O1	4.17
939182	AE1-148 E O1	2.78
939371	AE1-168 C	5.23
939372	AE1-168 E	3.48
939411	AE1-173 C	37.39
939412	AE1-173 E	24.92
AA2-074	AA2-074	2.63
CARR	CARR	0.31
CBM-S1	CBM-S1	6.65
CBM-S2	CBM-S2	7.55
CBM-W1	CBM-W1	6.62
CBM-W2	CBM-W2	43.29
CIN	CIN	3.0
CPL	CPL	3.87
G-007	G-007	1.05
IPL	IPL	1.88
LGEE	LGEE	0.87
MEC	MEC	6.69
MECS	MECS	2.77
O-066	O-066	6.69
RENSSELAER	RENSSELAER	0.24
WEC	WEC	0.8

13.10 Index 9

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
171935	314285	6CHRL249	DVP	314316	6LOCKS	DVP	1	DVP_P4-2: 562T563	breaker	684.0	126.64	127.73	AC	8.71

Bus #	Bus	MW Impact
314541	3WATKINS	0.23
314554	3BTLEBRO	0.4
314557	3BETHEL	0.39
314572	3EMPORIA	0.33
314574	6EVERETS	1.77
314578	3HORNRTN	1.92
314582	3KELFORD	1.62
314603	3SCOT NK	1.61
314617	3TUNIS	0.36
314620	6CASHIE	0.31
314623	3WITAKRS	0.69
314704	3LAWRENC	0.2
315102	1BRUNSWICKG1	3.87
315103	1BRUNSWICKG2	3.87
315104	1BRUNSWICKG3	3.87
315105	1BRUNSWICKS1	8.03
315131	1EDGECEMA	4.73
315132	1EDGECEMB	4.73
315136	1ROSEMG1	1.22
315137	1ROSEMS1	0.76
315138	1ROSEMG2	0.57
315139	1GASTONA	1.77
315141	1GASTONB	1.77
315150	1BUGGS 1	5.14
315151	1BUGGS 2	5.14
900672	V4-068 E	0.13
916301	Z1-086 C	23.58
916302	Z1-086 E	5.22
917332	Z2-043 E	0.46
917342	Z2-044 E	0.3
917512	Z2-088 E OP1	1.76
918492	AA1-063AE OP	2.11
918512	AA1-065 E OP	1.81
918532	AA1-067 E	0.31
918562	AA1-072 E	0.08
919692	AA2-053 E	2.07
919702	AA2-057 E	1.83
920042	AA2-088 E OP	4.27
920592	AA2-165 E	0.24
920672	AA2-174 E	0.24
923851	AB2-025 C	0.41

Bus #	Bus	MW Impact
923852	AB2-025 E	1.3
923911	AB2-031 C O1	1.6
923912	AB2-031 E O1	0.79
923991	AB2-040 C O1	0.74
923992	AB2-040 E O1	4.29
924022	AB2-043 E O1	2.32
924152	AB2-059 E O1	2.75
924162	AB2-060 E O1	1.9
924302	AB2-077 E O1	0.6
924312	AB2-078 E O1	0.6
924322	AB2-079 E O1	0.6
924401	AB2-089 C	1.07
924402	AB2-089 E	0.55
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.25
924502	AB2-099 E	0.11
924511	AB2-100 C	10.62
924512	AB2-100 E	5.23
925122	AB2-169 E	2.2
925171	AB2-174 C O1	5.2
925172	AB2-174 E O1	4.7
925591	AC1-034 C	3.46
925592	AC1-034 E	2.61
925612	AC1-036 E	0.6
925781	AC1-054 C O1	3.69
925782	AC1-054 E O1	1.7
926071	AC1-086 C	14.05
926072	AC1-086 E	6.39
926201	AC1-098 C	3.11
926202	AC1-098 E	1.86
926211	AC1-099 C	1.04
926212	AC1-099 E	0.61
926271	AC1-105 C O1	2.36
926272	AC1-105 E O1	1.18
927021	AC1-189 C	3.85
927022	AC1-189 E	1.92
927141	AC1-208 C	4.79
927142	AC1-208 E	2.13
927251	AC1-221 C	0.93
927252	AC1-221 E	0.93
927261	AC1-222 C	1.47
927262	AC1-222 E	1.4
930402	AB1-081 E O1	1.94
930861	AB1-132 C O1	1.35
930862	AB1-132 E O1	4.09
931231	AB1-173 C	0.23
931232	AB1-173 E	0.75
931241	AB1-173AC	0.23
931242	AB1-173AE	0.75
932631	AC2-084 C	4.44
932632	AC2-084 E	2.19

Bus #	Bus	MW Impact
932761	AC2-100 C	2.13
932762	AC2-100 E	1.04
933991	AD1-023 C	4.64
933992	AD1-023 E	2.53
934201	AD1-047 C	5.72
934202	AD1-047 E	3.81
934231	AD1-050 C	2.36
934232	AD1-050 E	1.29
934311	AD1-055 C	1.02
934312	AD1-055 E	0.26
934331	AD1-057 C O1	5.86
934332	AD1-057 E O1	3.12
934341	AD1-058 C	2.33
934342	AD1-058 E	0.59
934521	AD1-076 C	18.15
934522	AD1-076 E	9.24
934611	AD1-087 C O1	3.67
934612	AD1-087 E O1	1.72
934621	AD1-088 C	4.84
934622	AD1-088 E	2.27
934991	AD1-131 C	0.76
934992	AD1-131 E	0.51
935171	AD1-152 C O1	3.65
935172	AD1-152 E O1	2.43
935212	AD1-156 E	0.63
936261	AD2-033 C	5.16
936262	AD2-033 E	3.44
936331	AD2-043 C	1.91
936332	AD2-043 E	2.26
936361	AD2-046 C O1	4.24
936362	AD2-046 E O1	1.95
936401	AD2-051 C O1	3.84
936402	AD2-051 E O1	1.65
936481	AD2-063 C O1	6.36
936482	AD2-063 E O1	4.24
936531	AD2-068 C	2.27
936532	AD2-068 E	1.17
936701	AD2-089 C	3.47
936702	AD2-089 E	2.31
936711	AD2-090 C O1	2.58
936712	AD2-090 E O1	1.72
937481	AD2-202 C O1	1.01
937482	AD2-202 E O1	0.51
937571	AD2-169 C	6.85
937572	AD2-169 E	4.56
938171	AE1-026 C1 O	10.2
938172	AE1-026 C2 O	1.57
938173	AE1-026 E O1	2.99
938221	AE1-035 C	0.92
938222	AE1-035 E	0.45
938491	AE1-068 C O1	33.31
938492	AE1-068 E O1	18.4

Bus #	Bus	MW Impact
938501	AE1-069 C O1	26.32
938502	AE1-069 E O1	15.05
938661	AE1-088	1.26
939181	AE1-148 C O1	4.17
939182	AE1-148 E O1	2.78
939371	AE1-168 C	5.23
939372	AE1-168 E	3.48
939411	AE1-173 C	37.39
939412	AE1-173 E	24.92
AA2-074	AA2-074	2.63
CARR	CARR	0.31
CBM-S1	CBM-S1	6.65
CBM-S2	CBM-S2	7.55
CBM-W1	CBM-W1	6.62
CBM-W2	CBM-W2	43.29
CIN	CIN	3.0
CPL	CPL	3.87
G-007	G-007	1.05
IPL	IPL	1.88
LGEE	LGEE	0.87
MEC	MEC	6.69
MECS	MECS	2.77
O-066	O-066	6.69
RENSSELAER	RENSSELAER	0.24
WEC	WEC	0.8

13.11 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172864	314697	6SEEDGE HILL	DVP	927250	AC1-221 TAP	DVP	1	DVP_P1-2: LN 556	single	674.92	104.77	107.9	AC	20.89

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.46
314704	3LAWRENC	0.16
315150	1BUGGS 1	17.2
315151	1BUGGS 2	17.2
315153	1CLOVER1	32.32
315154	1CLOVER2	31.9
315156	1HALLBR1	1.05
315158	1KERR 1	0.26
315159	1KERR 2	1.05
315160	1KERR 3	1.03
315161	1KERR 4	1.03
315162	1KERR 5	1.03
315163	1KERR 6	1.03
315164	1KERR 7	1.03
315165	1HURT 1	5.16
315166	1HURT 2	5.16
315170	1BREMO 3	6.17
315171	1BREMO 4	13.55
315191	1BEARGRDN G1	2.35
315192	1BEARGRDN G2	2.35
315193	1BEARGRDN S1	4.84
315266	1PLYWOOD A	2.73
920291	AA2-127	0.49
924021	AB2-043 C O1	0.6
924031	AB2-045 C	0.35
924161	AB2-060 C O1	1.73
924301	AB2-077 C O1	0.38
924311	AB2-078 C O1	0.38
924321	AB2-079 C O1	0.38
924401	AB2-089 C	1.61
925611	AC1-036 C	0.18
925661	AC1-042 C	1.59
925781	AC1-054 C O1	4.97
925831	AC1-062	0.06
925991	AC1-075 C	11.93
926021	AC1-080 C	3.99
926271	AC1-105 C O1	14.49
926451	AC1-116 C	0.04
926641	AC1-145 C	1.9
927261	AC1-222 C	10.12
932511	AC2-071 C	0.34

Bus #	Bus	MW Impact
933501	AC2-165 C	3.59
934231	AD1-050 C	3.55
934311	AD1-055 C	7.02
934611	AD1-087 C O1	26.93
934621	AD1-088 C	31.82
935171	AD1-152 C O1	26.77
935221	AD1-157 C	0.3
935231	AD1-160 C	1.55
936261	AD2-033 C	17.81
936331	AD2-043 C	13.11
936361	AD2-046 C O1	8.38
936481	AD2-063 C O1	20.35
937481	AD2-202 C O1	7.42
938371	AE1-056 C	9.12
938821	AE1-108 C O1	8.24
939181	AE1-148 C O1	8.48
939371	AE1-168 C	20.89
939941	AE1-230 C1	0.72
AC1-131	AC1-131	5.19
BLUEG	BLUEG	6.71
CALDERWOOD	CALDERWOOD	1.58
CANNELTON	CANNELTON	0.47
CATAWBA	CATAWBA	1.69
CBM-N	CBM-N	0.03
CHEOAH	CHEOAH	1.48
CHILHOWEE	CHILHOWEE	0.51
COFFEEN	COFFEEN	0.81
COTTONWOOD	COTTONWOOD	5.16
DUCKCREEK	DUCKCREEK	1.65
EDWARDS	EDWARDS	0.73
FARMERCITY	FARMERCITY	0.6
G-007A	G-007A	0.32
GIBSON	GIBSON	0.29
HAMLET	HAMLET	3.17
NEWTON	NEWTON	2.11
NYISO	NYISO	0.14
PRAIRIE	PRAIRIE	4.53
SANTEETLA	SANTEETLA	0.44
SMITHLAND	SMITHLAND	0.41
TATANKA	TATANKA	1.0
TILTON	TILTON	0.86
TRIMBLE	TRIMBLE	0.73
TVA	TVA	4.31
UNIONPOWER	UNIONPOWER	2.22
VFT	VFT	0.83

13.12 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172708	314747	6BREMO	DVP	313867	6BREMODIST	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	117.21	120.35	AC	20.88

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.5
314704	3LAWRENC	0.13
315150	1BUGGS 1	7.33
315151	1BUGGS 2	7.33
315153	1CLOVER1	6.11
315154	1CLOVER2	6.03
315158	1KERR 1	0.17
315159	1KERR 2	0.71
315160	1KERR 3	0.7
315161	1KERR 4	0.7
315162	1KERR 5	0.7
315163	1KERR 6	0.7
315164	1KERR 7	0.7
315170	1BREMO 3	30.25
315171	1BREMO 4	66.47
315191	1BEARGRDN G1	12.56
315192	1BEARGRDN G2	12.56
315193	1BEARGRDN S1	25.85
315266	1PLYWOOD A	0.57
920291	AA2-127	2.61
923911	AB2-031 C O1	0.67
923991	AB2-040 C O1	0.31
924021	AB2-043 C O1	0.36
924031	AB2-045 C	0.89
924161	AB2-060 C O1	1.06
924301	AB2-077 C O1	0.22
924311	AB2-078 C O1	0.22
924321	AB2-079 C O1	0.22
924401	AB2-089 C	1.15
925171	AB2-174 C O1	2.12
925611	AC1-036 C	0.16
925781	AC1-054 C O1	3.68
925831	AC1-062	0.07
925991	AC1-075 C	2.53
926021	AC1-080 C	0.85
926271	AC1-105 C O1	3.21
927251	AC1-221 C	0.76
927261	AC1-222 C	1.83
931231	AB1-173 C	0.1
931241	AB1-173AC	0.1
932511	AC2-071 C	0.88

Bus #	Bus	MW Impact
932761	AC2-100 C	1.74
933501	AC2-165 C	32.69
934201	AD1-047 C	2.4
934231	AD1-050 C	2.53
934311	AD1-055 C	1.27
934341	AD1-058 C	1.9
934611	AD1-087 C O1	4.25
934621	AD1-088 C	14.59
934991	AD1-131 C	0.62
935171	AD1-152 C O1	4.23
935221	AD1-157 C	0.34
935231	AD1-160 C	1.78
936261	AD2-033 C	14.42
936331	AD2-043 C	2.38
936361	AD2-046 C O1	5.5
936481	AD2-063 C O1	14.24
937481	AD2-202 C O1	1.17
938371	AE1-056 C	10.47
938821	AE1-108 C O1	38.04
939181	AE1-148 C O1	5.53
939371	AE1-168 C	20.88
939841	AE1-220 C O1	12.08
AA2-074	AA2-074	1.48
CARR	CARR	0.29
CBM-S1	CBM-S1	2.82
CBM-S2	CBM-S2	4.08
CBM-W1	CBM-W1	1.69
CBM-W2	CBM-W2	17.19
CIN	CIN	0.85
CPLE	CPLE	2.17
IPL	IPL	0.51
LGEE	LGEE	0.25
MEC	MEC	2.24
MECS	MECS	0.24
RENSSELAER	RENSSELAER	0.23
WEC	WEC	0.22

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172786	314765	6MTEAGLE	DVP	314749	6CHARLVL	DVP	1	DVP_P1-2: LN 2027-B	single	661.76	111.79	114.93	AC	20.88

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.5
314704	3LAWRENC	0.13
315150	1BUGGS 1	7.33
315151	1BUGGS 2	7.33
315153	1CLOVER1	6.11
315154	1CLOVER2	6.03
315158	1KERR 1	0.17
315159	1KERR 2	0.71
315160	1KERR 3	0.7
315161	1KERR 4	0.7
315162	1KERR 5	0.7
315163	1KERR 6	0.7
315164	1KERR 7	0.7
315170	1BREMO 3	30.25
315171	1BREMO 4	66.47
315191	1BEARGRDN G1	12.56
315192	1BEARGRDN G2	12.56
315193	1BEARGRDN S1	25.85
315266	1PLYWOOD A	0.57
920291	AA2-127	2.61
923911	AB2-031 C O1	0.67
923991	AB2-040 C O1	0.31
924021	AB2-043 C O1	0.36
924031	AB2-045 C	0.89
924161	AB2-060 C O1	1.06
924301	AB2-077 C O1	0.22
924311	AB2-078 C O1	0.22
924321	AB2-079 C O1	0.22
924401	AB2-089 C	1.15
925171	AB2-174 C O1	2.12
925611	AC1-036 C	0.16
925781	AC1-054 C O1	3.68
925831	AC1-062	0.07
925991	AC1-075 C	2.53
926021	AC1-080 C	0.84
926271	AC1-105 C O1	3.2
926451	AC1-116 C	0.52
927251	AC1-221 C	0.76
927261	AC1-222 C	1.83
931231	AB1-173 C	0.1
931241	AB1-173AC	0.1

Bus #	Bus	MW Impact
932511	AC2-071 C	0.88
932761	AC2-100 C	1.74
933501	AC2-165 C	32.68
934201	AD1-047 C	2.4
934231	AD1-050 C	2.53
934311	AD1-055 C	1.27
934341	AD1-058 C	1.9
934611	AD1-087 C O1	4.25
934621	AD1-088 C	14.59
934991	AD1-131 C	0.62
935171	AD1-152 C O1	4.23
935221	AD1-157 C	0.34
935231	AD1-160 C	1.78
936261	AD2-033 C	14.42
936331	AD2-043 C	2.38
936361	AD2-046 C O1	5.5
936481	AD2-063 C O1	14.24
937481	AD2-202 C O1	1.17
938371	AE1-056 C	10.47
938821	AE1-108 C O1	38.04
939181	AE1-148 C O1	5.53
939371	AE1-168 C	20.88
939841	AE1-220 C O1	12.08
AA2-074	AA2-074	1.48
CARR	CARR	0.29
CBM-S1	CBM-S1	2.81
CBM-S2	CBM-S2	4.07
CBM-W1	CBM-W1	1.68
CBM-W2	CBM-W2	17.14
CIN	CIN	0.84
CPL	CPL	2.17
IPL	IPL	0.51
LGEE	LGEE	0.25
MEC	MEC	2.23
MECS	MECS	0.23
RENSSELAER	RENSSELAER	0.23
WEC	WEC	0.22

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
172765	927250	AC1-221 TAP	DVP	304070	6PERSON230 T	CPL	1	DVP_P1-2: LN 556	single	718.0	108.19	111.08	AC	20.89

Bus #	Bus	MW Impact
314429	3JTRSVLE	0.46
314704	3LAWRENC	0.16
315150	1BUGGS 1	17.2
315151	1BUGGS 2	17.2
315153	1CLOVER1	32.32
315154	1CLOVER2	31.9
315156	1HALLBR1	1.05
315158	1KERR 1	0.26
315159	1KERR 2	1.05
315160	1KERR 3	1.03
315161	1KERR 4	1.03
315162	1KERR 5	1.03
315163	1KERR 6	1.03
315164	1KERR 7	1.03
315165	1HURT 1	5.16
315166	1HURT 2	5.16
315170	1BREMO 3	6.17
315171	1BREMO 4	13.55
315191	1BEARGRDN G1	2.35
315192	1BEARGRDN G2	2.35
315193	1BEARGRDN S1	4.84
315266	1PLYWOOD A	2.73
920291	AA2-127	0.49
924021	AB2-043 C O1	0.6
924031	AB2-045 C	0.35
924161	AB2-060 C O1	1.73
924301	AB2-077 C O1	0.38
924311	AB2-078 C O1	0.38
924321	AB2-079 C O1	0.38
924401	AB2-089 C	1.61
925611	AC1-036 C	0.18
925661	AC1-042 C	1.59
925781	AC1-054 C O1	4.97
925831	AC1-062	0.06
925991	AC1-075 C	11.93
926021	AC1-080 C	3.99
926271	AC1-105 C O1	14.49
926451	AC1-116 C	0.04
926641	AC1-145 C	1.9
927251	AC1-221 C	11.73
927261	AC1-222 C	10.12

Bus #	Bus	MW Impact
932511	AC2-071 C	0.34
932761	AC2-100 C	26.99
933501	AC2-165 C	3.59
934231	AD1-050 C	3.55
934311	AD1-055 C	7.02
934341	AD1-058 C	29.4
934611	AD1-087 C O1	26.93
934621	AD1-088 C	31.82
934991	AD1-131 C	9.64
935171	AD1-152 C O1	26.77
935221	AD1-157 C	0.3
935231	AD1-160 C	1.55
936261	AD2-033 C	17.81
936331	AD2-043 C	13.11
936361	AD2-046 C O1	8.38
936481	AD2-063 C O1	20.35
937481	AD2-202 C O1	7.42
938371	AE1-056 C	9.12
938821	AE1-108 C O1	8.24
939181	AE1-148 C O1	8.48
939371	AE1-168 C	20.89
939941	AE1-230 C1	0.72
AC1-131	AC1-131	5.19
BLUEG	BLUEG	6.71
CALDERWOOD	CALDERWOOD	1.58
CANNELTON	CANNELTON	0.47
CATAWBA	CATAWBA	1.69
CBM-N	CBM-N	0.03
CHEOAH	CHEOAH	1.48
CHILHOWEE	CHILHOWEE	0.51
COFFEEN	COFFEEN	0.81
COTTONWOOD	COTTONWOOD	5.16
DUCKCREEK	DUCKCREEK	1.65
EDWARDS	EDWARDS	0.73
FARMERCITY	FARMERCITY	0.6
G-007A	G-007A	0.32
GIBSON	GIBSON	0.29
HAMLET	HAMLET	3.17
NEWTON	NEWTON	2.11
NYISO	NYISO	0.14
PRAIRIE	PRAIRIE	4.53
SANTEETLA	SANTEETLA	0.44
SMITHLAND	SMITHLAND	0.41
TATANKA	TATANKA	1.0
TILTON	TILTON	0.86
TRIMBLE	TRIMBLE	0.73
TVA	TVA	4.31
UNIONPOWER	UNIONPOWER	2.22
VFT	VFT	0.83

Affected Systems

14 Affected Systems

14.1 Duke Energy Progress

None.

Short Circuit

15 Short Circuit

The following Breakers are overduty:

None

Stability Study

16 Stability

To be completed as part of the Facilities Study

Attachment 1 Single Line Diagram

